

Learning Plan

This task helps connect you with the unit, its assessments, and the planning necessary to achieve your target grade outcomes.

Show us that you can:

- Plan out how you will engage with the unit, with dates highlighted in the learning plan timeline.
- Reflect on your current knowledge and your goals in the unit.

Complete the following:

1. Download the template from the OnTrack resources - review the template carefully and complete the responses.
2. Upload to OnTrack once ready. Login to OnTrack and mark the task as **Ready for Feedback**. Upload the required files and reflect on what you learned.
3. Engage with feedback received from your tutor.

To get the task Complete, you need to engage with the feedback you receive. Your tutor will review your submission and may ask you to clarify aspects of the task or include aspects you have missed.

Remember, a 'Resubmit' status is not a bad thing; your tutor is just trying to help improve your understanding and your results. You have not lost any marks. Work with your tutor and your final portfolio will achieve a higher final score as a result. In this way, a 'Resubmit' is a good thing!

When you do resubmit, make sure to include a comment that describes how you have addressed the feedback you received. This needs to demonstrate how you have addressed all the aspects indicated by your tutor in their feedback on your learning.

Tip: Engage with your tutors!

To succeed in this unit you must engage with your tutors. Interact with them during seminars and on the SIT102 Teams channel. Read and respond to feedback received on your OnTrack submissions.

Task in Detail

Programming is foundational to computer and data science, artificial intelligence, cyber security, and software engineering. Through this unit you will learn how computers work and how to get them to do anything you want. Your success here ensures that you will have the knowledge and skills to succeed throughout your future studies. This makes introductory programming the most important building block in any computing degree, and a valuable skill for any profession.

In this unit you can determine which grade you are aiming to achieve, and we will work with you to help you achieve it.

Focus and Unit Learning Goals

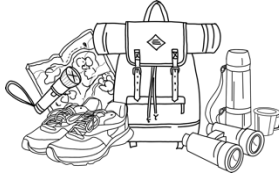
To complete this unit, you need to demonstrate that you can do the following:

- ULO 1: Evaluate procedural program code for correct use of coding conventions and use code tracing and debugging techniques to identify and correct issues.
- ULO 2: Design, develop, and test procedural programs using specified programming languages to achieve defined program goals, including effective use of data types, programming statements, control flow structures and modularisation techniques.
- ULO 3: Explain the principles of structured procedural programming, using appropriate terminology and by relating these principles to programming syntax and structures developed.
- ULO 4: Analyse, critique the quality, and reflect upon a portfolio of artefacts to justify the achievements of specified objectives and goals with evidence.

Through the unit you will create a portfolio of tasks that shows you have these capabilities and reflect on your achievement of these at the end of the unit.

Meaning of grades

Different grades require you to demonstrate different skills.

			
Pass	Credit	Distinction	High Distinction
Can work through structured resources to iteratively develop understanding. Has a foundational understanding of programming.	Can effectively utilise feedback to develop their understanding and pays attention to detail to ensure solutions work as expected. Has a good understanding of programming, and how to program.	Can work with little direction, with the persistence and adaptability needed to overcome challenges in building larger solutions. Has the capability to design and build programs independently.	Can work in a proactive, self-directed manner, and demonstrates a degree of passion or inventiveness in their work. Has the capability to conceptualise, design and build elegant and sophisticated programs.

When you complete all the tasks up to your target grade, you know that you have demonstrated the required skills and will receive the indicated grade at the end of the unit. *You can change your target grade at any point.*

The meaning of each grade is described below. It relates to the type of tasks you will be working on at that grade level. It also indicates how you should approach the unit and manage your time if targeting that grade.

Pass: Has a foundational understanding of programming

Through the **pass tasks** you will develop an understanding of the basic programming tools and how to apply them to simple problems.



If you aim for a pass, you will be demonstrating that you can apply these tools and concepts to create small programs but **not** that you are able to create more complex programs yourself or fully understand how programs work.

If you choose to target a pass grade, you will have more time to spend on the pass tasks as you will not have to complete other tasks. However, achieving only a pass may make future computing units more challenging. Building a solid foundation here will be important for you.

Make sure you let your tutor know why you have chosen to target this grade. If the reason is that you believe you will struggle with this unit, they will support your learning as best they can, and maybe, even change your target and achieve a credit or higher.

Credit: Has a good understanding of programming, and how to program

The **credit tasks** will ask you to think deeply about the concepts learned and how they relate to each other, so you really understand how things work. This will give you the opportunity to put into practice the concepts learned and ensure that you really know how they work together.

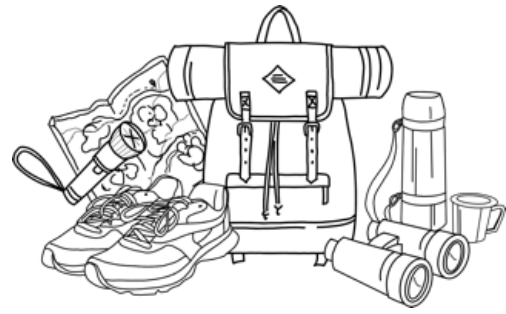


If you aim for credit, you will be demonstrating that you understand how programs work, and that you can draw the necessary parts together to build programs with some guidance.

If you choose to target a credit grade, you will have more time to spend on the pass and credit tasks as you will not have to complete higher-grade tasks. Make sure you let your tutor know why you have chosen to target this grade so they can provide you with the support you need.

Distinction: Has the capability to create programs independently

The **distinction tasks** will ask you to create larger programs and give you the freedom to determine what these programs do. You will scope, design, code, and test these programs, working independently to create programs to meet specified requirements.

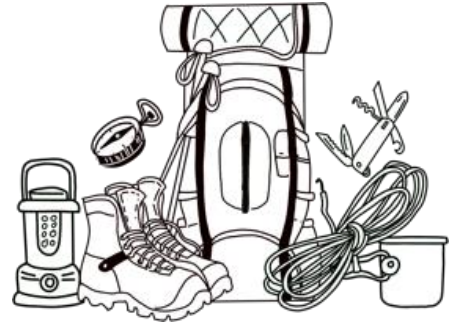


If you aim for a distinction, you will be demonstrating that you understand how programs work and can independently design and develop a program.

If you choose to target a distinction grade, you will have less time to spend on the pass and credit tasks as you will need that learning to tackle the distinction tasks. Make sure you keep track of your time and plan ahead to ensure you can complete all the required tasks. Let your tutor know why you have chosen to target this grade so they can provide you with the support you need.

High distinction: Has the capability to build elegant and sophisticated programs.

The **high distinction tasks** will ask you to demonstrate high levels of programming competency in your custom program, and your



ability to incorporate concepts and tools you have learnt independently. The instructions are open-ended so you have lots of freedom on what you can do.

If you aim for a high distinction, you will be demonstrating that you understand what programming can do, can independently find resources to support programming efforts, and are growing in your capability to build large, elegant, and sophisticated software solutions.

If you choose to target a high distinction grade, you will have less time to spend on the pass and credit tasks as you will need that learning to tackle the higher-grade tasks. Make sure to communicate with your tutor about your plans for the high-distinction task and discuss it with them.

Feedback

Feedback provides you guidance on how to improve. With OnTrack, you receive feedback from your tutor for each task submitted **before** its due date. Tutor feedback will include if there are things that you need to address or questions they have about your understanding. When you address the feedback, the task will be marked as Complete.

Getting the tasks marked as Complete in OnTrack is your way of knowing you will get the grade you are aiming for. If all the tasks are Complete – you already know your grade before submitting your portfolio!

You are encouraged to submit tasks early. This is important because you may need to take feedback in and resubmit.

Showing that you can receive and respond to feedback will help improve your final grade, and this process will help you learn. Ensure you make the most of the feedback dialogue and don't be scared to ask questions!

Learning with others

Learning with others in this unit is not only promoted but expected. Learning with peers through asking for help, supporting others, or sharing your learning, has many benefits for all involved, such as social support and encouragement.

However, you need to be careful to avoid **collusion** when working with others!

What is collusion? Copying or letting someone copy your work – *even if it is in small amounts*. For that reason, when you help others or ask help from others, ensure the conversation focuses on explanations of how things work, how to do things, or how to think about addressing the task, rather than providing answers.

What is plagiarism? You are free to use learning resources to *assist* your learning. This includes the Programmers.Guide, online resources, and even AI tools like ChatGPT. **However**, all submitted work must be entirely your own. You cannot copy or use any code directly – they are only to be used to help you understand topics, and *not* to complete work for you. Evidence of submitted code that has been copied from external sources is a very serious offence, and will result in penalties such as failing the task or even failing the entire unit. If ever in doubt, please speak with your tutors, and browse Deakin's [AI-Use Guides](#).

Timelines

Tutors will regularly provide feedback and check tasks. We aim to provide you with feedback within five business/working days. Again, you only receive feedback if the task is submitted by its due date.

If you are aiming for a higher grade, *you need to work ahead of these times* to fit in the subsequent tasks. The following table provides **suggested** submission dates. You can use this to help plan your time.

Task		Suggested Week Due by Grade			
		Pass	Credit	Distinction	HD
P1	Learning Plan	1	1	1	1
P2	Building and Running	1	1	1	1
P3	Sequence and Data	2	2	2	2
P4	Control Flow	4	4	3	3
P5	Progress checkpoint	4	4	4	4
T1	Test 1 (run in week 5)	5	5	5	5
P6	C / C++	5	5	4	4
P7	Structuring code	7	6	5	4
P8	Structuring data	8	7	5	5
P9	Working with multiples	9	8	6	5
D2	Custom Program Plan	-	-	6	6
T2	Test 2 (run in week 9, 10, and 11) - hurdle to pass subject	9	9	9	9
C1	Indirect Access	-	9	8	7
C2	Another Language - Python	-	10	9	8
D3	Deep Dive Memory	-	-	9	8
C3	Classes and Objects	-	11	10	9
D4	Custom Program Implementation	-	-	11	10
D1	Active Collaborator	-	-	11	11
H1	Something Awesome	-	-	-	11
P10	Learning Summary	11	11	11	11

Test 1 and Test 2

There are two tests that you need to sit as part of your portfolio. Test 1 occurs in week 5, and Test 2 occurs between weeks 9 and 11. These are required learning activities in your timetable, so ensure you are allocated appropriately.

Campus Students

If you are enrolled on campus for your course, you will sit these tests at the indicated location in your timetable. The test will be conducted on paper (so bring along a pen/pencil). No other material will be permitted.

Online Students

You will sit an online version. Afterwards, Test 2 requires a short interview with a tutor in which you must demonstrate an adequate level of understanding for the task to be marked as complete. Details on this process will be communicated to all students.

For online interviews, you need to ensure you have a working microphone and camera, and adequate network bandwidth for clear communication.

Don't Stress

Test 1 and 2 are only checking that you have a good understanding of the essential content. If you are progressing through the tasks, you should have no trouble completing them. *You must pass Test 2 to pass the unit*, but there are opportunities to re-sit if you really miss something important, or to fix the test when you get it close enough.

We promise that there will not be trick questions, there will be plenty of time, and that the practice tests provided will be very close to the real test. Our goal is to validate that you have achieved the required understanding to pass.

Furthermore, your tutors will do live demonstrations of practice questions during the seminars. Attendance to seminars is absolutely key, as students who attend seminars regularly often find Test 1 and Test 2 quite manageable.

How to Succeed in SIT102:

Attend seminars: The seminars provide a great environment to engage with the unit, clarify difficulties with your tutors and peers, and accelerate your progress.

Students who attend seminars regularly find the OnTrack tasks easier to complete, stay up-to-date, and generally find the unit quite manageable.

Students who miss seminars tend to take much longer to complete OnTrack tasks, find them more difficult to complete, and find themselves to be less knowledgeable about key topics. Each year ~25% of students fail the unit – the majority of these are students who never (or rarely) engaged with their tutors.

Engage and be involved: successful students are those that ask their tutors questions, **are active on the SIT102 Teams channel**, and reach out for help when they need. You'll learn a lot from these interactions, so please make yourself known to your tutors and classmates.

Stay up-to-date with your tasks: we have OnTrack tasks due each week, so it helps to complete tasks in a timely fashion so you don't have a large collection of tasks to complete all at once. During seminars can be a great time to complete your OnTrack tasks.

Have fun! Programming can seem challenging at first, but it is important to know that you are not the only one who feels that way. The majority of SIT102 students are brand new to programming (and also new to University). Please know that we have lots of support available (seminars, SIT102 Teams channel, tutors, the Unit Chair) so if you are ever unsure, please don't be embarrassed to reach out for assistance. You may also consider joining some excellent [Deakin Clubs](#) to meet fellow students and find a welcoming community.